

A High-Sensitivity 200 Gbps PAM4 Receiver with Flip-Chip Integrated Si-Ge Avalanche Photodiode and CMOS Transimpedance Amplifier

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Abstract This paper reports a high-sensitivity optical receiver based on flip-chip integrated Si-Ge avalanche photodiode and CMOS transimpedance amplifier. Clear eye diagrams are achieved at 100 Gbps NRZ and 200 Gbps PAM4, with sensitivities of -26.1 dBm and -20 dBm, respectively.

Introduction

With the rapid growth of data-intensive applications such as artificial intelligence (AI) and high-performance computing (HPC), the demand for high-speed, high-sensitivity optical receivers in data center interconnects continues to increase. Current 400G/800G optical modules widely adopt PAM4 modulation, with per-lane data rates scaling toward 200 Gbps. However, the accumulated insertion losses introduced by high-density integration, wavelength-division multiplexing (WDM), and advanced packaging make it increasingly challenging to maintain sufficient link margin under constrained optical power budgets. Therefore, enhancing receiver sensitivity while preserving high bandwidth has become a critical challenge in short-reach optical interconnects [1].

Avalanche photodiodes (APDs) provide internal gain via impact ionization, enabling improved receiver sensitivity and relaxed noise requirements for the transimpedance amplifier (TIA) compared to PIN PDs, and thus have attracted significant interest for high-speed optical receivers [2, 3]. However, the performance of APDs in high-speed operation is fundamentally limited by the gain-bandwidth product (GBP), which is closely related to the ionization coefficient ratio k in the multiplication region. Conventional InGaAs/InP APDs offer excellent sensitivity but require high operating voltages and exhibit poor compatibility with standard complementary metal-oxide-semiconductor (CMOS) processes, leading to increased fabrication cost and limited scalability for low-cost, large-volume integration [4].

Silicon-germanium (Si-Ge) APDs employ Ge as the absorption layer and Si as the multiplication layer, combining high near-infrared absorption efficiency with low-noise avalanche characteristics enabled by the low ionization coefficient ratio k of Si. In addition, Si-Ge technology is highly com-

patible with standard CMOS platforms, offering advantages in cost and integration density, and has attracted considerable attention in recent years [5, 6]. However, despite demonstrated high device performance, system-level validation in integrated receiver front-ends remains limited [7, 8], leaving a gap between device performance and practical applications.

In this work, we demonstrate a flip-chip (FC) integrated Si-Ge APD-TIA optical receiver, with the APD fabricated on a 65 nm silicon-on-insulator (SOI) platform and the TIA in a 28 nm CMOS process, achieving an energy efficiency of 93.5 fJ/bit at 200 Gbps. By optimizing the multiplication and charge control regions in the separate-absorption-charge-multiplication (SACM) structure, the APD achieves a maximum avalanche gain of 95 at a reverse bias voltage of 14.5 V. The FC integration effectively reduces parasitic capacitance and inductance between the APD and TIA, enabling a 3 dB optoelectronic (OE) bandwidth of 42 GHz. At the system level, the integrated receiver achieves direct detection of 100 Gbps NRZ and 200 Gbps PAM4 signals without digital signal processing (DSP), with clear and open eye diagrams. At the soft-decision forward error correction (SD-FEC) threshold, the receiver sensitivities are -30.2/-26.1 dBm for 64/100 Gbps NRZ and -23.1/-20 dBm

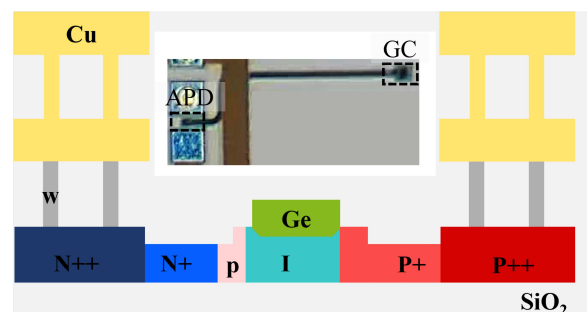


Fig. 1: Cross-sectional schematic of the Si-Ge APD; inset: micrograph of the fabricated device.

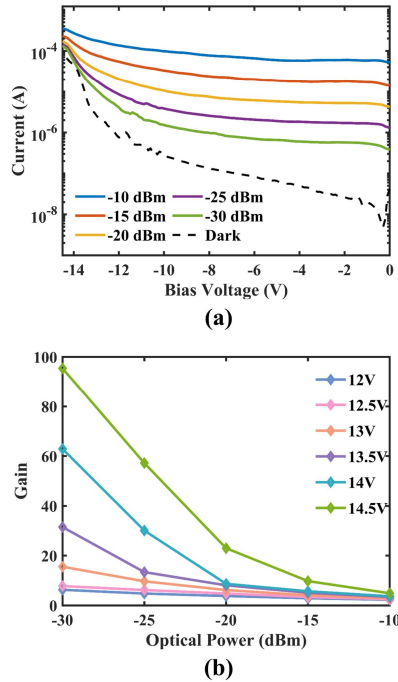


Fig. 2: (a) Measured IV curves with different input optical powers; (b) The dependence of gain on input optical power across various bias voltages.

for 100/200 Gbps PAM4, respectively. These results demonstrate a favorable trade-off between bandwidth and sensitivity, highlighting the potential of the proposed Si-Ge APD-TIA receiver for next-generation short-reach optical interconnects.

Device Design

Fig. 1 shows the cross-sectional schematic of the device. The Si-Ge APD used in this work is fabricated on a silicon photonics platform and adopts a separate absorption, charge, and multiplication (SACM) structure, where Ge serves as the absorption layer and Si as the multiplication layer. The inset of Fig. 1 presents a micrograph of the fabricated device. Light is coupled into the underlying Si waveguide via a grating coupler (GC) and subsequently evanescently coupled into the Ge absorption region for photodetection. Photons absorbed in the Ge layer generate electron-hole pairs, which drift under the electric field into the Si multiplication region, where impact ionization occurs, leading to avalanche gain.

In the SACM structure, the absorption and multiplication regions are spatially separated: a relatively low electric field is maintained in the absorption region to suppress premature avalanche, while a high electric field is established in the multiplication region to enable efficient impact ionization. In addition, a charge layer is introduced to precisely tailor the electric field distribution, confining the high field to the multiplication region. This design ensures stable and efficient avalanche multiplication while preventing premature breakdown

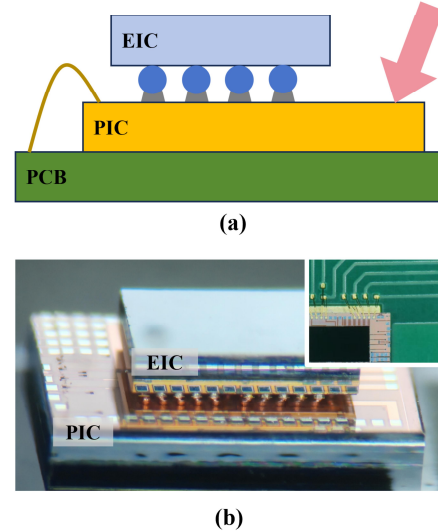


Fig. 3: (a) Schematic of FC integration; (b) Micrograph of the integrated receiver, with the inset showing a top view.

in the absorption region.

APD Static Characterization

Fig. 2(a) shows the current-voltage (I-V) characteristics of the fabricated waveguide Si-Ge APD under different reverse bias, measured under both dark conditions and optical input powers ranging from -30 dBm to -10 dBm. By comparing with a reference p-i-n PD on the same chip, the unity-gain responsivity is determined to be 0.8 A/W at a reverse bias of 2 V. As the reverse bias increases, the dark current gradually rises to 9.4 μ A at $\sim$ 13.5 V, remaining sufficiently low for high receiver sensitivity. The breakdown voltage is measured to be 14.5 V, defined as the bias at which the dark current reaches 100 μ A [9].

Fig. 2(b) shows the avalanche gain as a function of reverse bias. The avalanche multiplication gain $G(V)$ is defined as $G(V) = (I_p(V) - I_d(V)) / (I_p(V_0) - I_d(V_0))$ where $I_p(V)$ and $I_d(V)$ denote the photocurrent and dark current at bias V , respectively, and V_0 is the unity-gain reference bias (2 V in this work). As the bias increases from 12 V toward breakdown, the gain increases monotonically, reaching 31, 62, and 95 at reverse biases of 13.5 V, 14 V, and 14.5 V, respectively.

Flip-Chip Integration

To realize high-speed and low-parasitic integration of the APD and TIA, FC bonding is employed, as shown in Fig. 3(a). First, an under-bump metallization (UBM) layer is formed on the pads of the photonic integrated circuit (PIC) to serve as the foundation for subsequent micro-bump formation. Micro-bumps are then fabricated via electroplating, followed by FC alignment and bonding of the electronic integrated circuit (EIC) onto the PIC, enabling high-density electrical interconnection. In addition, wire bonding is used to connect

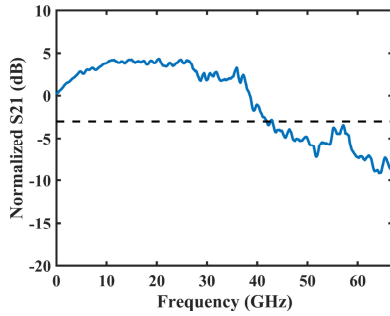


Fig. 4: Measured frequency response of the receiver.

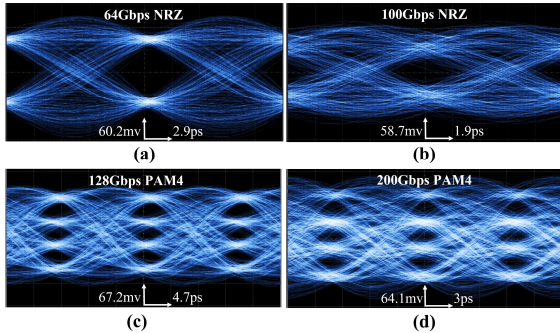


Fig. 5: Measured eye diagrams at data rates of (a) 64 Gbps NRZ, (b) 100 Gbps NRZ, (c) 128 Gbps PAM4, and (d) 200 Gbps PAM4.

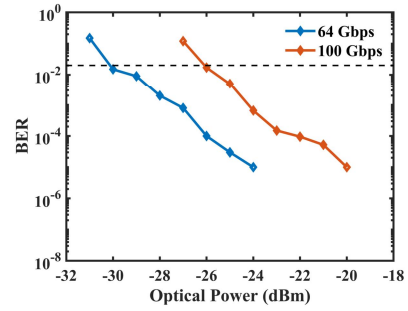
the ground pads of the PIC to the printed circuit board (PCB) for external interfacing.

A micrograph of the integrated device is shown in Fig. 3(b), with the inset presenting a side view of the assembled receiver. The single-end TIA is fabricated in a 28-nm CMOS process with a transimpedance gain of $60 \text{ dB}\Omega$ ($1 \text{ k}\Omega$). The APD and TIA are interconnected through metal bumps, reducing interconnect inductance and parasitic capacitance, thereby improving high-frequency performance. This hybrid integration scheme combines the low-parasitic advantage of FC bonding with the flexibility of wire bonding, enabling high-speed operation and supporting 200 Gbps PAM4 signaling. The integrated receiver consumption of 18.7 mW, corresponding to an energy efficiency of 93.5 fJ/bit at 200 Gbps, making it well suited for power-constrained data center interconnects.

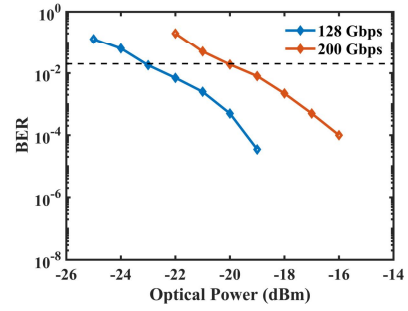
Receiver Dynamic Characterization

The small-signal frequency response of the integrated receiver was measured using a 67 GHz vector network analyzer (VNA), with light coupled via a grating coupler (GC). As shown in Fig. 4, a 3-dB bandwidth of 42 GHz is obtained, ensuring sufficient bandwidth for high-speed signal reception.

High-speed eye diagram measurements were performed to assess signal integrity. The optical setup includes an O-band laser, a polarization controller (PC), and a thin-film lithium niobate (TFLN) Mach-Zehnder modulator (MZM) driven by a Keysight M8199B arbitrary waveform generator



(a)



(b)

Fig. 6: Measured BER versus received optical power for (a) NRZ and (b) PAM4 signals.

(AWG) and an electrical amplifier. The modulated optical signal is coupled into the device under test (DUT) via a grating coupler (GC).

Eye diagrams were measured at a reverse bias of 13.5 V and an input optical power of -15 dBm. As shown in Fig. 5, clear and open eyes are obtained for 64/100 Gbps NRZ and 128/200 Gbps PAM4 signals. The well-opened eyes indicate effective high-speed signal recovery. The ability to support 200 Gbps PAM4 operation, despite the 42 GHz bandwidth, is attributed to the use of a 5-tap feed-forward equalizer (FFE) in the oscilloscope, together with transmitter-side pre-emphasis enabled by the AWG calibration.

The bit-error-rates (BERs) performance was evaluated at the same bias, as shown in Fig. 6. At the SD-FEC threshold, receiver sensitivities of -30.2/-26.1 dBm are achieved for 64/100 Gbps NRZ, and -23.1/-20 dBm for 128/200 Gbps PAM4.

Conclusion

We demonstrate a high-speed optical receiver based on FC integrated Si-Ge APD and TIA for direct 200 Gbps PAM4 detection without DSP. The SACM APD achieves a gain of 95 at 14.5 V. With reduced parasitics enabled by FC integration, the receiver achieves a 42 GHz bandwidth and -20 dBm sensitivity at 200 Gbps PAM4. These results confirm that Si-Ge APDs enable simultaneous high bandwidth and high sensitivity, and that advanced packaging provides a viable solution for 200 Gbps per wavelength short-reach optical interconnects.

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